

IMPLEMENTING A SIMPLE NEURAL NETWORK ON THE TITANIC DATASET

1. INTRODUCTION

The Titanic dataset is one of the most well-known datasets in data science and machine learning. It contains details of passengers aboard the RMS Titanic and their survival status. This case study aims to build a simple neural network model to predict whether a passenger survived the disaster based on available features.

2. PROBLEM STATEMENT

The objective is to classify passengers as either survivors or non-survivors using a neural network model. Given the dataset, we will preprocess the data, train a neural network, and evaluate its performance.

3. DATASET INFORMATION

- **Source:** "<https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv>"
- **Key Features:**
 - PassengerId (Unique ID of the passenger)
 - Pclass (Ticket class: 1 = First, 2 = Second, 3 = Third)
 - Name (Passenger's name)
 - Sex (Male/Female)
 - Age (Age in years)
 - SibSp (Number of siblings/spouses aboard)
 - Parch (Number of parents/children aboard)
 - Ticket (Ticket number)
 - Fare (Ticket fare)
 - Cabin (Cabin number)
 - Embarked (Port of embarkation: C = Cherbourg, Q = Queenstown, S = Southampton)
 - Survived (Target variable: 0 = No, 1 = Yes)

4. DATA PREPROCESSING

- **Handling Missing Values**
- **Encoding Categorical Data**
- **Feature Scaling**
- **Splitting Data**

5. MODEL IMPLEMENTATION

- **Define the Neural Network Architecture:**
 - Input layer with necessary features.
 - Hidden layers with activation functions (e.g., ReLU).
 - Output layer with a sigmoid activation function for binary classification
- **Compile the Model:**
 - Use binary cross-entropy as the loss function.
 - Use an optimizer such as Adam
- **Train the Model:**
 - Train on the training dataset with appropriate batch size and epochs.
 - Monitor validation performance to avoid overfitting.
- **Evaluate the Model:**
 - Assess accuracy, precision, recall, and F1-score on the test dataset.
 - Use confusion matrix and ROC curve for performance analysis

6. CONCLUSION

- Summarize key findings and model performance.
- Discuss potential improvements, such as hyperparameter tuning and feature selection.

- Suggest further enhancements, like using more complex deep learning architectures or ensemble models.

